

H.W: Lab 1

1. Find the frequency counts for the following

i) for ($i=1; i \leq n; i++$);
 printf("Hello");

ii) $i=1;$
 $i = i+20;$
 printf("%d", i);

iii) $i=1;$
 while ($i \leq n$)
 {
 printf("%d", i); $i++$;
 }

For all the shell programs done in OS Lab. Find out time complexity for all the program.

Statement	Cost	Repetition	Total
for ($i=0; i \leq n; i++$);	1+1+1	1+(n+1)+n	2n+2
printf("Hello");	1	1	1
Total			2n+3

Statement	cost	Repetition	Total
$i = 1$;	1	1	1
$i = i + 20$;	2	1	2
$\text{printf}("%d", i);$	1	1	1
Total.			4

Statement	cost	Repetition	Total
$i = 1$;	1	1	1
$\text{while}(i \leq n)$	1	$n+1$	$n+1$
$\text{printf}("%d", i);$	1	n	n
$i++$;	1	n	n

$3n+2$

2) Basic shell program:

```
#!/bin/bash
```

```
echo -n "Enter the first number:"
```

```
read num1.
```

```
echo -n "Enter the second number:"
```

```
read num2.
```

```
sum=$((num1+num2))
```

```
echo "The sum of $num1 and $num2 is:
```

```
$sum"
```


code	cost	Repetition	Total
echo -n "Enter the first number"	1	1	1
read num1	1	1	1
echo -n "Enter the second number"	1	1	1
read num2	1	1	1
Sum = $\$((num1 + num2))$	2	1	2
echo -n "The sum of \$num1 and \$num2 is: " \$sum	1	1	1
			7

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